

Multiscale computation of plume expansion in thin atmospheres

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The effects of ambient rarefaction on plume flow expansion from a continuum to a rarefied regime are poorly understood. The ambiguity in selecting appropriate governing equations to minimize the computational cost and, at the same time, accurately predict the multiscale dynamics of such flows has led most previous studies to focus on the two extremes: vacuum plumes (where the ambient is vacuum) or continuum plumes (where the ambient is in the continuum regime). This study considers plume expansion under ambient conditions with various degrees of rarefaction, spanning continuum, transitional, and near-vacuum conditions. We achieve this by adapting the Unified Gas Kinetic Wave-Particle (UGKWP) method to a curvilinear grid framework for axisymmetric plume simulations. The method solves the Bhatnagar–Gross–Krook (BGK) collision model of the Boltzmann equation by coupling grid- and particle-based approaches without explicitly discretizing the velocity space, offering reasonable computational cost across various rarefied regimes. Simulation results will be validated against existing continuum and near-vacuum plume flow data from the literature. The study is expected to provide a detailed understanding of the flow physics governing plume characteristics in transitional ambient conditions and scaling parameters to quantify changes in plume dynamics across various rarefied regimes.

I. Introduction

We are entering a new era of space exploration with NASA’s Artemis program aimed at landing humans on the Moon after over half a century. Unlike the Apollo program, the goal of the Artemis missions is to establish a long-term human presence on the Moon aimed at sending the first astronauts to Mars and making humans a multi-planetary species [1]. However, the extraterrestrial infrastructure needed to sustain a human presence requires landing payloads (≥ 20 metric tons) that are much larger than anything attempted before. Retrorockets that fire thrusters in the direction of flight to decelerate a vehicle have been identified as the enabling technology for descent and soft landing of the expected large payloads [2]. The efficacy of retrorockets in Earth’s atmosphere has been demonstrated multiple times by the SpaceX Falcon 9 first-stage booster recovery [3]. However, their success on Moon/Mars largely depends on accounting for the thin atmosphere of Moon/Mars [4].

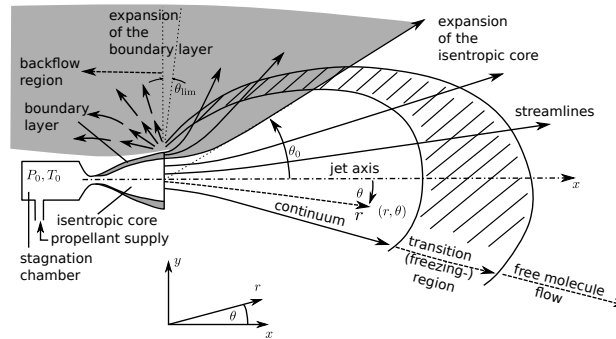


Fig. 1 Schematic of vacuum plume (adapted from [5])

Retrorocket exhaust expansion in low ambient pressure (or vacuum/rarefied) environments generates a plume-like flow field known as the vacuum plume [6]. A schematic of such plume expanding into vacuum like atmosphere is given in fig 1. The vacuum plume consists of inviscid, isentropic core and nozzle wall boundary layer, both of which

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expand into the thin external atmosphere. The rapid expansion of the viscous flow within the boundary layer into the low-pressure external atmosphere results in plume backflow. Furthermore, the flow at the nozzle exit is in the continuum regime and transitions to the free molecular regime farther downstream, making the flow field multiscale in nature. Due to such complex nature of plume expansion, the effects of atmospheric rarefaction on the plume flow, especially at hypersonic/supersonic speeds, are not well understood, and hence their influence on the aerodynamic forces and moments experienced by the spacecraft cannot be accurately estimated. Moreover, the plume backflow may cause re-impingement on the rocket body, leading to high heat fluxes and contamination on the rocket surface, undesired force/torques [7, 8] and damage to rocket sensors [9].

Several experiments were conducted from the 1960s to the 1990s with the broader goal of developing simple empirical models to describe plume flow behavior in vacuum environment. Early studies focused on the isentropic core flow of the plume, assuming inviscid conditions [10–15]. These studies were later extended to include viscous effects in order to characterize the expansion of the boundary layer and the consequent backflow [16–20], gas properties [21, 22], and boundary layer thickness [23, 24]. Bird [25] identified the breakdown of translational and rotational equilibrium in such highly expanded flows in rarefied environment due to the departure of the flow from continuum regime. Subsequent researches focused on the effects of this rarefaction on flow properties such as density, temperature, and Mach number [26]. The impact of rarefaction on flow angle [27, 28] and the resulting contamination due to species separation [29] were also investigated. Furthermore, plume surface interaction is a major concern during various phases of spacecraft operations like rendezvous, docking, maneuvers, etc., and also in the landing and takeoff of lunar vehicles. A thorough assessment of the plume surface interaction was made by [30, 31] among others in vacuum atmosphere. A detailed summary of these studies is provided by Dettleff in his review paper [5]. These researches were focused towards characterizing the plume flow at near-vacuum conditions while providing a theoretical, numerical and experimental basis for the governing laws that need to be used in those near-vacuum conditions. The phenomena governing change in plume properties for the transition region as the ambient condition is gradually rarefied starting from continuum is relatively unexplored. The effects of such rarefaction were briefly discussed by Muntz *et al.* [32], where they highlighted its effects on shock thickness variation in a plume flow with increase in rarefaction. They introduced a rarefaction variable ξ on which shock thickness of a given plume size depends. Along those lines, our goal is to identify scaling parameter(s) that may help collapse the centerline velocity, pressure, and density profiles across continuum to transition regimes.

The experimental assessment of plumes under such varying ambient conditions requires a vacuum chamber that can produce continuum to transition flows. Despite extensive efforts, conducting such tests remains challenging because of the issues and high cost of maintaining near-vacuum conditions. Furthermore, the study over a particular parameter (say one value of $\frac{P_0}{P_\infty}$) until the flow reaches a steady state is challenging because of the difficulty associated with sustaining high pressure in upstream pressurized tank or vacuum condition in downstream vacuum chamber type setup. As a result, the experiments conducted and the measurements/data made available to date are insufficient for a comprehensive assessment of plume expansion and related non-equilibrium phenomena in rarefied and even transition regime [33, 34].

Numerical simulations offer a viable alternative for investigating plume flows in rarefied conditions. Existing numerical methods to solve for flows spanning continuum to rarefied regimes include the Direct Simulation Monte Carlo (DSMC) [35] method that solves the Boltzmann equation in the whole domain and the hybrid multiscale methods that solve the Navier-Stokes (NS) equations in the continuum regimes and the Boltzmann equation in the rarefied regimes, with buffer zones coupling the two regions [36]. The DSMC approach uses operator splitting method to treat convection and collision of particles as separate processes. Due to this treatment, the numerical timestep (Δt) in DSMC must be lower than mean collision time between molecules. This makes the method prohibitively expensive in the high-density regions [35, 37] where mean collision time is very low. On the other hand, there is no straightforward procedure in the hybrid approaches to specify the buffer zones. Location of the interface between the solvers is usually identified using continuum breakdown by a parameter called gradient-length local Knudsen Number (Kn_{GL}) [35]. Kn_{GL} of 0.05 to 0.1, depending on the flow case, is usually used as a boundary line between the two solvers [38]. However, typical contour plots of the parameter for a multiscale flow show that constant Kn_{GL} region is not a smooth line, but a fluctuating one [39], and ad-hoc technique like image-processing is applied over the Kn_{GL} contour plot to facilitate domain classification. For instance in [40], the authors use a 5x5 median filter to smooth out the fluctuating lines and extract a clean curve corresponding to the constant Kn_{GL} region. Using such ad-hoc method may lead to inaccuracies in the buffer zone location, and this can introduce significant non-physical artifacts in the numerical solutions [41].

In this study, we address some of the critical gaps in understanding plume dynamics within transition regime. In particular, we plan to study the change in plume centerline velocity, pressure and density profiles with the onset of ambient rarefaction starting from continuum regime. Our major focus will be on the transition regime, and we aim to formulate a scaling parameter that may collapse these profiles under various rarefactions. Furthermore, the study

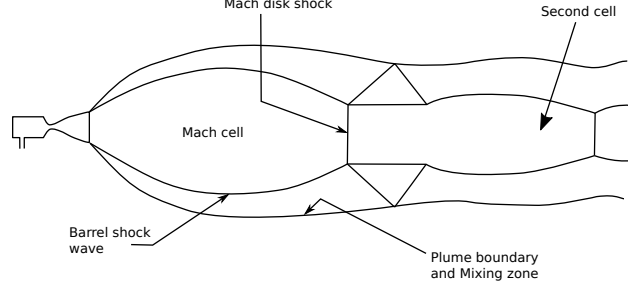


Fig. 2 Schematic of plume field in near-continuum regime (adapted from [32])

will systematically investigate changes in the shock structure pattern for plume flow under the combination of different ambient and nozzle exit conditions. This will be characterized with the change in shape and structure of mach-cell (Fig. 2) as well as shock thickness with increase in rarefaction, particularly for the transition regime. Moreover, the effects of rocket descent velocity (retro-propulsion effects) to such plume flow structures will also be studied. These objectives will be achieved through the development of a cost-efficient, high-fidelity multiscale solver, under a unified framework, capable of simulating flows across all regimes—from continuum to rarefied. In particular, we use the Unified Gas Kinetic Wave-Particle (UGKWP) method [42, 43], a unified computational method based on the Bhatnagar-Gross-Krook (BGK) [44] model of the Boltzmann equation. The multiscale nature of the method allows adaptive switching between NS-type gas kinetic scheme and DSMC-type particle method based on local cell Knudsen number (Kn_c). The accuracy of the solver is rigorously validated for continuum to near-continuum flows against established in-house continuum jet flow simulations [45] and past experimental results on vacuum plume [28, 46–50]. The major parameters for the study are nozzle exit Mach number (M_e) and nozzle exit to free-stream pressure ratio ($\frac{P_e}{P_\infty}$) that results in varying freestream Knudsen numbers (Kn_{ref}).

II. Methodology

A. Simulation Domain

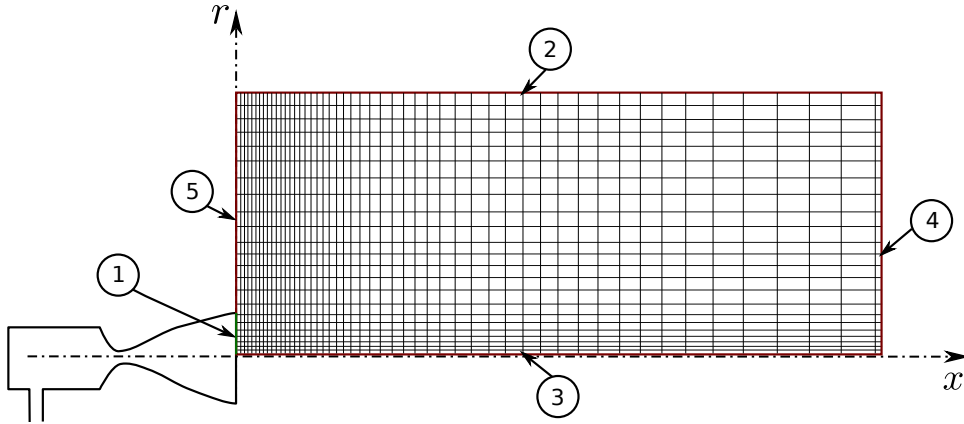


Fig. 3 Schematic of domain and mesh for numerical simulation

The grid used for the simulation is stretched towards the nozzle exit in x -direction and towards the axis in r -direction as shown in Fig. 3. Different types of boundaries for the numerical simulation are given in Table. 1. The mesh used requires transforming the governing equations so that they confirm to stretched coordinate system (x, r) . In the current work, BGK model for Boltzmann equation, transformed to curvilinear coordinate system (ξ, η) given by [51] is used for simulation. A schematic showing such grid transformation is given in Fig. 4. The curvilinear transformation is more

Table 1 Boundaries for numerical simulation

Boundary	Boundary type
1	Inlet
2, 4, 5	Outflow
3	Axisymmetry

general than required for the current orthogonal stretched grid, and the implementation will allow simulation over more complex surfaces in the future. The following sections describe the governing equations and their transformation in brief.

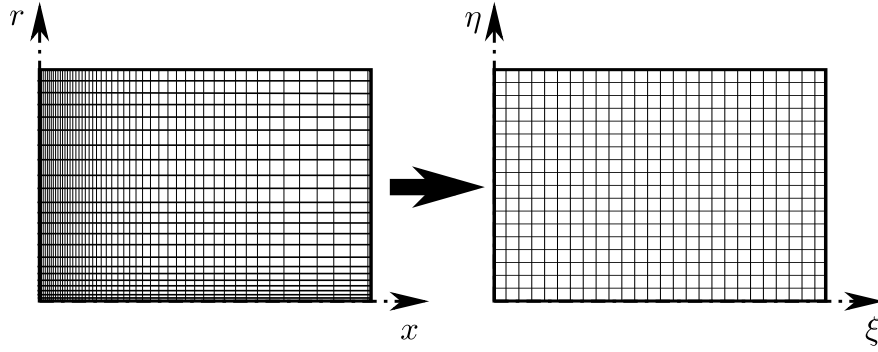


Fig. 4 Physical and numerical domain

B. BGK model

The BGK model for the Boltzmann equation is given by [44]:

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_{\vec{x}} f = \frac{g - f}{\tau}, \quad (1)$$

where f is velocity distribution function and $\vec{v}$ is particle velocity. g is a maxwellian (equilibrium) distribution for particle velocities and has following form:

$$g = \rho \left(\frac{\lambda}{\pi} \right)^{\frac{N+3}{2}} e^{-\lambda[(v-V)^2 + \xi^2]} \quad (2)$$

where $\lambda = m/2k_B T$, k_B is Boltzmann constant and T is macroscopic temperature. τ is the relaxation time and has form of [52]

$$\tau = \frac{\mu}{P} \quad (3)$$

where μ is viscosity and P is pressure.

The exact time-dependent velocity distribution function obtained by solving the model Eq. (1) along particle trajectories is represented by [53]:

$$f(\vec{x}, t, \vec{v}) = \frac{1}{\tau} \int_0^t e^{-(t-t')/\tau} g(\vec{x}', t', \vec{v}) dt' + e^{-t/\tau} f_0(\vec{x}_0, \vec{v}), \quad (4)$$

where $(\bullet_0)$ denote quantities at initial time ($t = 0$) and $(\bullet')$ denotes quantities along particle trajectory. Equation (4) can be rewritten as [42]:

$$f(\vec{x}, t, \vec{v}) = (1 - e^{-t/\tau}) g^+(\vec{x}, t, \vec{v}) + e^{-t/\tau} f_0(\vec{x}_0, \vec{v}), \quad (5)$$

with,

$$g^+(\vec{x}, t, \vec{v}) = g(\vec{x}, t, \vec{v}) + \left(\frac{te^{-t/\tau}}{1 - e^{-t/\tau}} - \tau \right) (\partial_t g(\vec{x}, t, \vec{v}) + \vec{v} \cdot \nabla_x g(\vec{x}, t, \vec{v})). \quad (6)$$

BGK formulation models collision process of the particles as a relaxation process which relaxes fluid particles towards equilibrium with a relaxation time τ . For a finite volume formulation using above governing equation, we are interested in flux construction at control volume faces using the exact solution of BGK equation given by (5). The exact solution at a face ($x = 0$ for convenience), describes how the distribution of particles at the face evolves over time. Initially, at $t = 0$, all particles at the face follow the initial gas distribution function, f_0 . This means that the particles at the face at $t = 0$ have not yet experienced any collisions. As time progresses, if we observe the same face and count the particles that remain collisionless, the number of particles still following the initial distribution, f_0 , decreases exponentially by a factor of $(e^{-\frac{t}{\tau}})$. Meanwhile, the number of particles in the face that have experienced at least one collision increases by a factor of $(1 - e^{-\frac{t}{\tau}})$ [54]. The particles that have undergone collisions follow the equilibrium distribution, g^+ [42].

C. UGKWP method

Using the numerical method, we aim to track how the distribution function evolves over a numerical timestep Δt , as determined by the Courant–Friedrichs–Lewy (CFL) condition. The first part of Eq. (5) corresponding to g^+ , that is, the time-dependent evolution of the equilibrium part, is fully determined by the right and left macroscopic states. These states, for a finite volume formulation, corresponds to macroscopic flow variables in right and left control volumes of a control surface (face). Consequently, the macroscopic flux contribution over the span of Δt due to g^+ can be determined by taking its moments and integrating over the time interval. However, for the second part, the initial distribution function at the face (f_0) needs to be first obtained. The process of obtaining (f_0) and its macroscopic flux contribution can be better understood by contrasting three different gas kinetic schemes: Gas Kinetic Scheme for Navier Stokes equations (GKS-NS) [53], Unified Gas Kinetic Scheme (UGKS) [55] and the method employed in the current paper, Unified Gas Kinetic Wave-Particle method (UGKWP) [42, 43].

In GKS-NS, Chapman-Enskog expansion [52] is used to approximate f_0 at a face as small perturbation from the maxwellian distributions in two neighboring cells. Hence, f_0 can be uniquely determined from macroscopic variables in left and right cells of the face. This, however, can only be applied to near-equilibrium flows due to the inherent assumptions made by Chapman-Enskog when calculating f_0 as a small perturbation from the equilibrium state g .

In UGKS, Finite Volume Method (FVM) is directly applied to evolve the discrete f at discrete velocity points in each cell. These discrete distribution functions at the cell centers are then reconstructed at the faces, at the start of each timestep, to obtain discrete f_0 's and ultimately summed up using discrete point integration method to take its moments and obtain macroscopic flux contribution. This approach, similar to Discrete Velocity Method (DVM), however, turns out to be very expensive in highly non-equilibrium flows because a large range of particle velocities — and thus a large number of velocity points are required to accurately capture all the non-equilibrium effects.

To reduce the computational cost, particles are introduced in the UGKWP method to account for the non-equilibrium effects instead of reconstructing discrete distribution functions at face and tracking their evolution over time. Particles are sampled in numerical cells (control volumes) using equilibrium distribution given by Eq. (6) at the start of each timestep Δt and their evolution for maximum time of Δt is governed by time exact solution of BGK model given in Eq. (5). Particles crossing the cell interface without undergoing even a single collision during Δt are counted. The equilibrium part in the flow is calculated similar to GKS or UGKS, while non-equilibrium part of the flow is calculated by summing up total properties transported by the particles leaving or entering the cell (similar to flux). The following sections briefly describe the formulation to account for equilibrium and non-equilibrium contributions.

1. Curvilinear formulation for Equilibrium calculation:

Taking moments of Eq. (1) in velocity space, the conservative form can be written as [53],

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} + \frac{\partial H}{\partial z} = 0, \quad (7)$$

where,

$$Q = \begin{pmatrix} \rho \\ \rho U \\ \rho V \\ \rho W \\ \rho E \end{pmatrix}; \begin{pmatrix} F \\ G \\ H \end{pmatrix} = \int \begin{pmatrix} u \\ v \\ w \end{pmatrix} \psi f d\Xi; \psi = \begin{pmatrix} 1 \\ u \\ v \\ w \\ \frac{1}{2}(u^2 + v^2 + w^2 + \zeta^2) \end{pmatrix} \quad (8)$$

Curvilinear formulation of Eq. (7) in arbitrary coordinates system (ξ, η, ζ) is given by [51].

$$\frac{\partial \hat{Q}}{\partial t} + \frac{\partial \hat{F}}{\partial \xi} + \frac{\partial \hat{G}}{\partial \eta} + \frac{\partial \hat{H}}{\partial \zeta} = 0, \quad (9)$$

where $\hat{Q} = \mathcal{J}Q$ and $\hat{F}$ is the flux in ξ direction given by

$$\hat{F} = \begin{cases} \hat{F}_\rho = S_\xi F_{\bar{\rho}}, \\ \hat{F}_{\rho u} = S_\xi (a_{11} F_{\bar{\rho} \bar{u}} + a_{12} F_{\bar{\rho} \bar{v}} + a_{13} F_{\bar{\rho} \bar{w}}), \\ \hat{F}_{\rho v} = S_\xi (a_{21} F_{\bar{\rho} \bar{u}} + a_{22} F_{\bar{\rho} \bar{v}} + a_{23} F_{\bar{\rho} \bar{w}}), \\ \hat{F}_{\rho w} = S_\xi (a_{31} F_{\bar{\rho} \bar{u}} + a_{32} F_{\bar{\rho} \bar{v}} + a_{33} F_{\bar{\rho} \bar{w}}), \\ \hat{F}_E = S_\xi F_{\bar{E}}. \end{cases} \quad (10)$$

where,

$$S_\xi = \sqrt{\hat{\xi}_x^2 + \hat{\xi}_y^2 + \hat{\xi}_z^2} \quad (11)$$

The flux expressions for η and ζ directions can be found in [51]. $\mathcal{J}$ is the Jacobian of transformation, and the metric terms have following form.

$$\begin{pmatrix} \hat{\xi}_x & \hat{\xi}_y & \hat{\xi}_z \\ \hat{\eta}_x & \hat{\eta}_y & \hat{\eta}_z \\ \hat{\zeta}_x & \hat{\zeta}_y & \hat{\zeta}_z \end{pmatrix} = \begin{pmatrix} y_\eta z_\zeta - z_\eta y_\zeta & z_\eta x_\zeta - x_\eta z_\zeta & x_\eta y_\zeta - y_\eta x_\zeta \\ y_\zeta z_\xi - z_\zeta y_\xi & z_\zeta x_\xi - x_\zeta z_\xi & x_\zeta y_\xi - y_\zeta x_\xi \\ y_\xi z_\eta - z_\xi y_\eta & z_\xi x_\eta - x_\xi z_\eta & x_\xi y_\eta - y_\xi x_\eta \end{pmatrix}. \quad (12)$$

The local particle velocities at the face, denoted in above formulation by $(\tilde{\bullet})$, are given by [51]

$$(\tilde{u}, \tilde{v}, \tilde{w}) = (u, v, w) \cdot (\hat{n}_x, \hat{n}_y, \hat{n}_z), \quad (13)$$

$$\hat{n}_x = \frac{(\hat{\xi}_x, \hat{\xi}_y, \hat{\xi}_z)}{\sqrt{\hat{\xi}_x^2 + \hat{\xi}_y^2 + \hat{\xi}_z^2}}, \quad (14)$$

$$\hat{n}_z = \frac{(x_\zeta, y_\zeta, z_\zeta)}{\sqrt{x_\zeta^2 + y_\zeta^2 + z_\zeta^2}}, \quad (15)$$

$$\hat{n}_y = \hat{n}_z \times \hat{n}_x. \quad (16)$$

Further details of the curvilinear formulation can be found in the reference [51]. A particular and important difference to note from the reference and in the current formulation is that the curvilinear implementation mentioned in this section is only used to calculate the equilibrium and wave part of the UGKWP method. Hence in the formulation mentioned above, (F, G, H) can be replaced by (F_{eq}, G_{eq}, H_{eq}) which corresponds to the term containing g^+ in Eq. (5) to obtain the contributions to the computational cell from the equilibrium fluxes. A similar approach is taken to account for wave fluxes that correspond to near-equilibrium particles.

2. Non-equilibrium calculation:

Particles are introduced at the beginning of timestep in each control volumes to account for non-equilibrium fluxes. Symmetric sampling method given by Bird [35] is applied to sample a pair of particles with velocities (u, v, w) and (u', v', w') from equilibrium distribution g as a first order approximation of g^+ . Here, g is defined in Eq. (2) and can

be constructed using solution in (ξ, η, ζ) coordinate system i.e. from $\hat{Q}^{n+1} (= \mathcal{J}Q^{n+1})$. These particle are moved in computational space (ξ, η, ζ) and their flux contribution to adjacent computational cells due to non-equilibrium transport can be accounted for.

Equal mass is assigned to each particle and is calculated from [43]:

$$m = \frac{\rho_h \times \Delta x}{N_p} \quad (17)$$

where ρ_h is density of collisional particles and N_p is some reference number of particles for each cell. A typical value of $N_p = 300$ is suitable for most cases and the number can be reduced for purely continuum or near-continuum flows.

The method reduces to a GKS based NS-type solver for small cell Knudsen number (Kn_c), where the dominant part of flux is from the equilibrium particles (called waves in the [43]). While for large (Kn_c), non-equilibrium particles (the particles that don't undergo collisions) are dominant, and the flux through the faces in the finite volume cell is mostly through this non-equilibrium transport of the particles from one cell to another. Hence, the flow variables in a cell are evolved over a CFL timestep Δt through the interplay of these wave and particle fluxes. Readers are suggested to go through [42, 43] for elaborate details of the method.

3. Update of solution

Solution at next timestep t^{n+1} can be obtained using explicit Euler method.

$$\begin{aligned} \hat{Q}_i^{n+1} = & \hat{Q}_i^n + \frac{1}{\Delta\xi} \left(\hat{F}_{\text{eq},i-1/2}^n - \hat{F}_{\text{eq},i+1/2}^n \right) + \frac{1}{\Delta\eta} \left(\hat{G}_{\text{eq},i-1/2}^n - \hat{G}_{\text{eq},i+1/2}^n \right) + \frac{1}{\Delta\zeta} \left(\hat{H}_{\text{eq},i-1/2}^n - \hat{H}_{\text{eq},i+1/2}^n \right) \\ & + \frac{1}{\Delta\xi} \left(\hat{F}_{\text{wv},i-1/2}^n - \hat{F}_{\text{wv},i+1/2}^n \right) + \frac{1}{\Delta\eta} \left(\hat{G}_{\text{wv},i-1/2}^n - \hat{G}_{\text{wv},i+1/2}^n \right) + \frac{1}{\Delta\zeta} \left(\hat{H}_{\text{wv},i-1/2}^n - \hat{H}_{\text{wv},i+1/2}^n \right) \\ & + \frac{1}{\Delta\xi\Delta\eta\Delta\zeta} \hat{\mathbf{w}}_{\text{fr}}. \end{aligned} \quad (18)$$

and these variables can be mapped back to physical space using

$$Q^{n+1} = \mathcal{J}^{-1} \hat{Q}^{n+1} \quad (19)$$

Here, term containing subscript ($\bullet_{\text{eq}}$) correspond to equilibrium fluxes through the cell interfaces and those containing subscript ($\bullet_{\text{wv}}$), called wave fluxes, are the non-equilibrium fluxes for near-equilibrium particles (particles that collide in the same numerical timestep Δt) modeled using macroscopic variables in neighboring cells. $\mathbf{w}_{\text{fr}}$ is flux due to free transport of the non-equilibrium (non-collisional) particles.

D. Boundary Conditions

Boundary condition (BC) implementation for UGKWP involves accounting for the BCs both in terms of equilibrium fluxes and particle fluxes. In the following section, boundary conditions needed for plume simulation is discussed in brief. All boundaries are treated in (ξ, η) space.

1. Inflow

For the plume flow in consideration, the inlet is supersonic and can be fully specified by incoming density, pressure (and consequently temperature) and velocity. The left-moving equilibrium flux may be ignored for supersonic inflow, but the UGKWP formulation adaptively assigns weight to the left and the right-moving fluxes at the face, so to avoid changing the formulation, both are specified. The flow variables in the cell center of the first cell are reconstructed with a biased upwind reconstruction using the first two cells. Hence, at the inlet boundary face left moving (exiting flux) is calculated using these reconstructed values of flow variables. For the right-moving equilibrium flux at the face (entering flux), this is calculated using boundary values (ρ, U, T) at the boundary face.

For particle fluxes, particles are generated from an equilibrium distribution corresponding to boundary conditions (ρ, U, T) . These particles are distributed uniformly in a ghost cell of volume $\Delta\xi\Delta\eta$ neighboring the boundary cells. Then, these particles are evolved over timestep Δt similar to the particles in the inner cells. The particles that enter the domain from boundary faces contribute to the non-equilibrium fluxes at the boundary.

2. Outflow

The outflow boundary condition for equilibrium part is applied with the assumption that the gradients in variables vanish at the outflow. To impose this, variables in inner cell left to the boundary cell is extrapolated with zero gradient to the boundary cell.

For the particles, those that leave the domain are deleted. If the outflow becomes subsonic or near sonic, there may be some particles entering the domain from the boundary as well. These particles are accounted for by extrapolating the boundary cell variables to a ghost cell and treating the particles in the ghost cell in a similar way as they are treated for inflow.

3. Axisymmetry

E. Validation Dataset

Validation will leverage the continuum simulation performed using NS compressible solver [45] and past experimental results on vacuum plume [28, 46–50]. Fig 5 shows density contours for an under-expanded Mach 1.5 jet with a nozzle exit to ambient pressure ratio ($\frac{P_e}{P_\infty}$) of 1.3 [45].

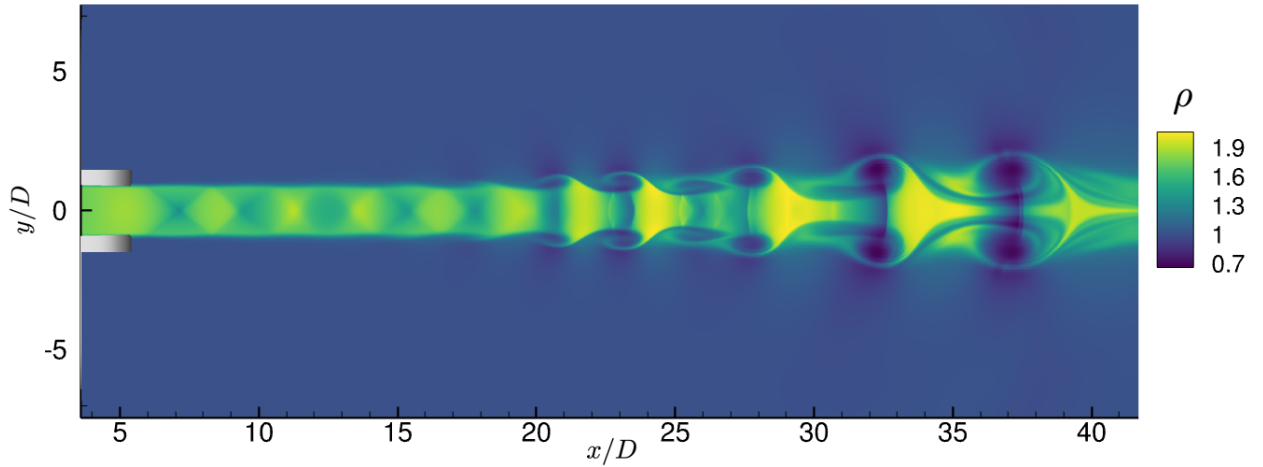


Fig. 5 Navier Stokes solution for continuum plume flow [45]

III. Numerical Validation

To date, the numerical solver for 1D problems and equilibrium part of 2D solver in the Cartesian coordinates has been successfully formulated and implemented. For validation, the 1D solver was tested for the 1D Sod shock tube test[42] case across a broad range of Knudsen numbers (from $\text{Kn} = 10^{-5}$ to $\text{Kn} = 10$) for a generic diatomic gas. The internal degrees of freedom (apart from translational in x) for the gas molecule are assumed to always remain in equilibrium. The results obtained were found to be consistent with reference solutions with reasonable accuracy. Further validation was conducted by solving for the internal profiles of normal shock structures at various Mach numbers for argon gas. Comparisons with DSMC solutions show good agreement in temperature and density profiles for normal shocks.

A. Sod shock tube

Numerical results from the developed 1-D solver for the Sod shock tests from reference Knudsen numbers (Kn_{ref}) ranging from 10^{-5} to 10 are shown in Fig. 6. For all the test cases, the total number of particles is maintained around 300 for each cell throughout the simulation. Dimensionless variables are used for simulation, and non-dimensionalization is performed using upstream variables. The computational domain is from 0 to 1 with 100 total cells and $\Delta x = 0.001$. Numerical timestep Δt is calculated using CFL conditions using the following relations:

$$\Delta t = \text{CFL} \frac{\Delta x}{\min(|U| + v)} \quad (20)$$

where v is the particle velocity.

Initial conditions for the case are as follows:

$$(\rho, U, P) = \begin{cases} (1.0, 0, 1.0) & x \leq 0.5, \\ (0.125, 0, 0.1) & x > 0.5 \end{cases}$$

The viscous coefficient is given as

$$\mu = \mu_{\text{ref}} \left(\frac{T}{T_0} \right)^\omega, \quad (21)$$

with the index for variable hard sphere (VHS) collision model, $\omega = 0.72$, and the reference viscosity [35]

$$\mu_{\text{ref}} = \frac{15\sqrt{\pi}}{2(5-2\omega)(7-2\omega)} \text{Kn}_{\text{ref}}. \quad (22)$$

For Kn_{ref} higher than 10^{-4} , ensemble averaging over 10 runs is performed to average out the fluctuations in the solution.

B. Normal Shock Structure

Numerical results for the internal profile of a Mach 8 normal shock are shown in Fig. 7. Results are calculated with approximately 300 particles in each cell. The domain extends from $x = -25$ to $x = 25$ with the initial discontinuity at $x = 0$, and is divided into 100 cells with $\Delta x = 0.001$. The same CFL condition given in Eq. (29) can be used to obtain the timestep Δt . The x-direction is non-dimensionalized using the initial upstream mean free path (mfp); hence, the domain is over a total of 50 mfp. The flow variables are non-dimensionalized using upstream variables. Rankine-Hugoniot normal shock relations for a Mach 8 shock are used to obtain the initial conditions.

Upstream (Left) Conditions:

$$\begin{aligned} \text{Ma}_l &= 8.0, \\ \rho_l &= 1.0, \\ U_l &= \text{Ma}_l \sqrt{\frac{\gamma}{2}}, \end{aligned} \quad (23)$$

Downstream (Right) Conditions:

$$\text{Ma}_r = \sqrt{\frac{\text{Ma}_l^2(\gamma - 1) + 2}{2\gamma\text{Ma}_l^2 - (\gamma - 1)}}, \quad (24)$$

$$\frac{T_r}{T_l} = \frac{\left(1 + \frac{\gamma-1}{2}\text{Ma}_l^2\right) \left(\frac{2\gamma}{\gamma-1}\text{Ma}_l^2 - 1\right)}{\text{Ma}_l^2 \left(\frac{2\gamma}{\gamma-1} + \frac{\gamma-1}{2}\right)}, \quad (25)$$

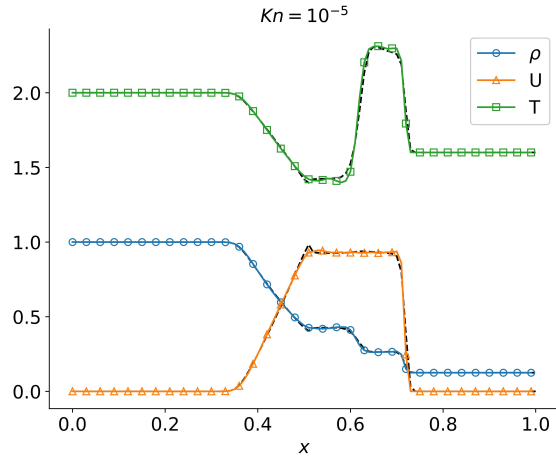
$$\rho_r = \rho_l \frac{(\gamma + 1)\text{Ma}_l^2}{(\gamma - 1)\text{Ma}_l^2 + 2}, \quad (26)$$

$$U_r = \text{Ma}_r \sqrt{\frac{\gamma}{2}} \sqrt{\frac{T_r}{T_l}}. \quad (27)$$

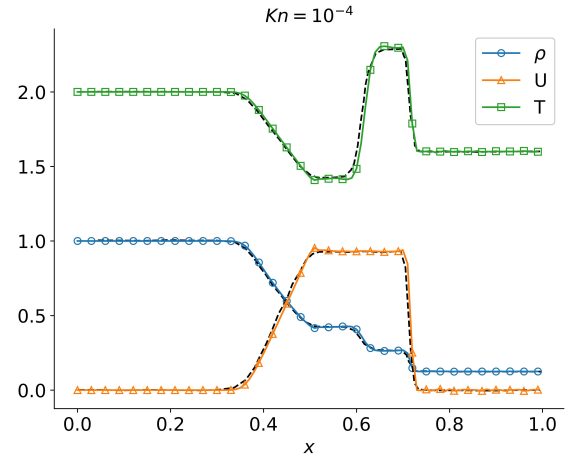
The gas used is monoatomic argon (Ar) gas with $\gamma = 1.66$ and follows the same viscosity laws as given in Eqs. (21) and (22) with VHS index of $\omega = 0.72$. Solution is time averaged over final one-fifth of total iterations.

IV. Discussion

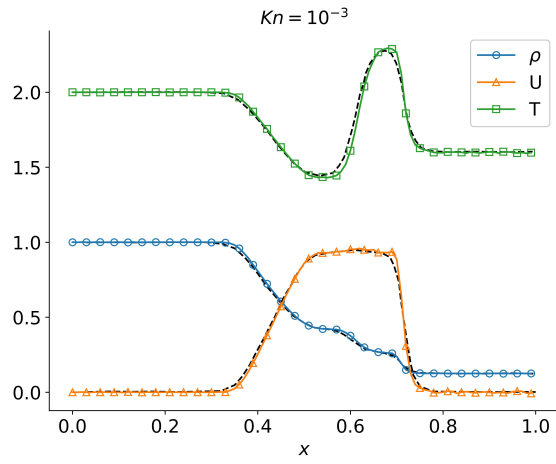
The developed UGKWP solver captures multiscale flow physics, particularly for the SOD test case with reasonable accuracy. In the test results (Fig. 6 (a to g)), the solver's performance across a wide range of Knudsen numbers (10^{-5} to 10) highlights its adaptability to both continuum and rarefied regimes. At low Kn (10^{-5} to 10^{-3}), on visual



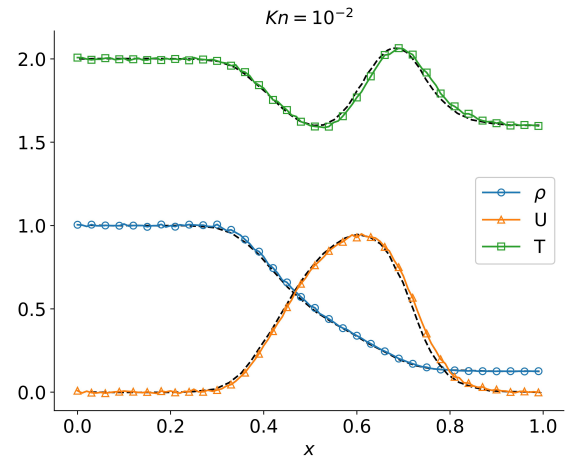
(6a)



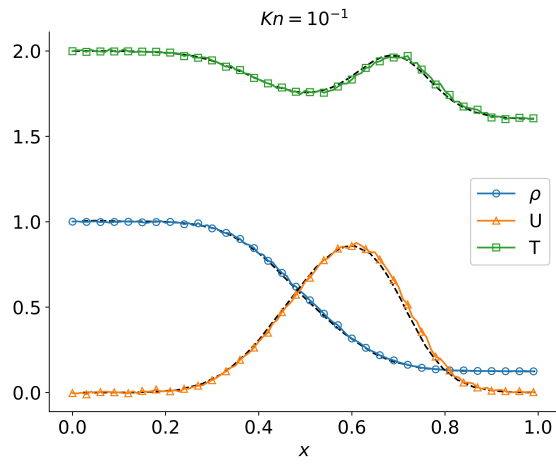
(6b)



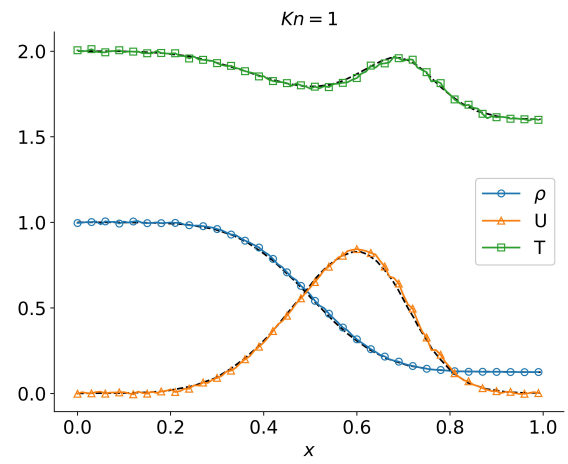
(6c)



(6d)



(6e)



(6f)

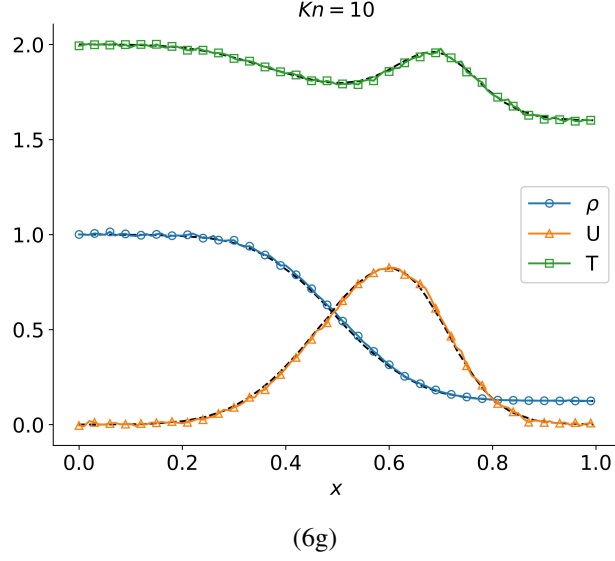


Fig. 6 Comparison of density, velocity, and equilibrium temperature profiles in the Sod shock tube problem at non-dimensional time $t = 0.12$. Subfigures (6a) to (6g) show results from $\text{Kn}_{\text{ref}} = 10^{-5}$ to $\text{Kn}_{\text{ref}} = 10$, both computed with spatial resolution $\Delta x = 0.01$ and CFL number 0.95. Dashed lines (- -) represent reference solutions from the Unified Gas Kinetic Scheme (UGKS), while solid lines (—) correspond to the current UGKWP method. Equilibrium quantities (ρ , U , p) are non-dimensionalized using the initial upstream equilibrium quantities. Plots demonstrate method's capability to solve for multiscale flows.

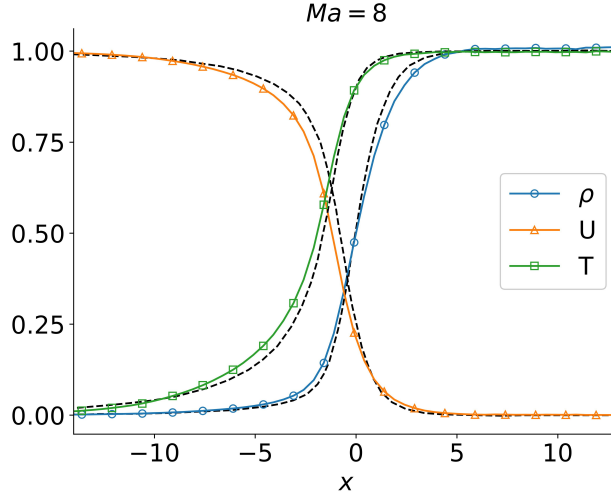


Fig. 7 Comparison of density, velocity, and temperature profiles for steady normal shock structure in Ar gas at $Ma = 8.0$. Results are computed with $\Delta x = 0.01$ and CFL number 0.95. Dashed lines (- -) show reference DSMC and solid lines (—) current UGKWP method results. x-axis is non-dimensionalized using initial upstream mean free path and all y-axis variables are normalized using the initial upstream properties obtained from Rankine-Hugoniot relations. The results demonstrate the scheme's accuracy in predicting highly non-equilibrium flows.

inspection, the results closely match UGKS reference solutions, with sharp shock and contact discontinuity resolutions similar to Navier-Stokes predictions. As Kn increases (10^{-2} to 10), progressive shock smearing is observed due to reduced interparticle collisions. In high- Kn flows, the dominance of non-equilibrium particle transport (where particles freestream multiple cells without collisions) leads to diffused shock fronts and broader contact regions. This behavior aligns with the physical expectation that rarefied flows exhibit weaker dissipation mechanisms, resulting in less distinct wave structures [26]. The UGKWP method inherently captures this transition by adaptively weighting wave (equilibrium) and particle (non-equilibrium) contributions. This outlines the method’s capability across different regimes without the need for explicit regime classification.

For the Mach 8 normal shock structure (Fig. 7), the internal density, velocity, and temperature profiles computed by UGKWP align closely with DSMC reference solutions, validating the solver’s ability to resolve highly non-equilibrium phenomena inherent to shock structures in rarefied environments. The gradual transition of flow variables across the shock, as opposed to sharp discontinuities, underscores the non-equilibrium effects arising from insufficient particle collisions in high-Mach-number, low-density flows. However, a slight offset in the shock location between UGKWP and DSMC results is observed on visual inspection, and this is due to the random walk of the particles. Specifically, the stochastic particle fluxes at subsonic outflow boundaries introduce fluctuation in the total number of particles in the flow domain and cause difficulties in maintaining a fixed shock position. This phenomenon is discussed by Bird in [35], and Bird suggests using a specularly reflecting wall boundary condition at the subsonic outlet to mitigate the displacement of the shock location. Even with this boundary condition, the shock location was found to be displaced by some factor, so he mentions a numerical stabilization to suppress the displacement. Using such a kind of numerical stabilization is non-physical but may ensure better results for the UGKWP similar to DSMC.

V. Conclusion and plan for the full paper

The developed method successfully demonstrates a unified framework for simulating 1D multiscale flows in rarefied conditions. The 1D solver, based on the UGKWP method, is validated against results from DSMC and UGKS methods and shows potential for simulating multiscale plume flows. The solver captures non-equilibrium physics dominant in the internal structure of Mach 8 shocks, though minor shock displacements are present due to the nature of implemented boundary conditions. Sod shock tube results show gradually increasing shock smearing with increasing Knudsen numbers, caused by reduced particle collisions in rarefied regimes. The observation aligns with physical expectations and demonstrates the capability of the solver in capturing multiscale flows.

Following 1-D validation, current work focuses on extension of the solver for 2-D flows. The 2-D solver will first be tested for cases with orthogonal mesh structures like 2-D SOD shock tube, lid-driven cavity flow, and Couette flow. Later on the solver will be extended for curvilinear grids and will be tested for hypersonic flow over a cylinder. To enhance computational efficiency, parallelization using MPI will be integrated for grids as well as particles. The above mentioned test cases will again be evaluated for parallel and curvilinear framework, and will be validated against reference data [42, 43]. Once the parallel solver is operational, the focus will shift to mesh generation and initial condition setup for actual plume flow simulations. The simulation results will be validated against available results given in Section II.E. A detailed analysis will be conducted to analyze the plume flow behavior with varying nozzle-exit conditions over a range of parameter space given in Table 2 and 3.

1. Parameter space for simulation in quiescent atmosphere

The major parameters for the experimental studies mentioned in section I are typically the total pressure to ambient pressure ratio ($\frac{P_0}{P_\infty}$) and the geometry of the nozzle to achieve the desired value of flow speed (or Mach number (M_e)) at the nozzle exit. For instance, in [30], the authors used a pressurized tank that supplied stagnation air at a total pressure (P_0) of 1.75×10^7 Pa and a total temperature (T_0) of 305K. Total pressure to ambient pressure ratio ($\frac{P_0}{P_\infty}$) was varied from 250×10^3 down to 50×10^3 . Two different nozzle geometries were used to achieve nozzle exit Mach numbers of 1 and 5. In [31], the study was made using total pressures of 2×10^5 , 8×10^5 and 16×10^5 Pa, and the pressurized gas was expanded to total pressure to ambient (pitot) pressure ratio ($\frac{P_0}{P_{t,\infty}}$) of 10^5 . One particular thing to note in all of these studies is the vacuum condition of ambient atmosphere. Dettleff, in his review [5], mentions that the pressure of atmosphere needs to be around 10^{-4} Pa for it to be recognized as true vacuum. For such true vacuum condition of 10^{-4} Pa and stagnation pressure of 1.75×10^7 Pa used by [30] in their experimental study, the ratio ($\frac{P_0}{P_\infty}$) turns out to be on the order of 10^{11} . Sustaining such high pressure ratio in experimental setup is not only difficult to achieve, but almost impossible. Hence, experimental studies using vacuum chambers, for instance [56], use vacuum chamber pressure on

the order of 10 Pa to characterize near-vacuum (rarefied) conditions.

Hence it can be deduced that stagnation pressure and temperature of the pressurized tank are the major parameters that can be varied in an experimental setup. To be consistent with experimental conditions, stagnation pressure and temperature of the tank are taken from [30].

Tank Stagnation Conditions and gas properties: $P_0 = 1.75 \times 10^7$ Pa ; $T_0 = 305$ K [30]. Gas in consideration for this study is an inert argon (Ar) gas with gas particle diameter of 4.17×10^{-10} m [35]. Variable hard sphere (VHS) collision is considered between gas particles and this corresponds to a VHS index (ω) of 0.72 [35]. Viscosity coefficient for the gas follows Eq. (21) and Eq. (22).

Nozzle exit conditions: Flow inside the nozzle is assumed to be isentropic to get nozzle exit static pressure at various nozzle exit Mach numbers (M_e). Ambient (free-stream) condition is assumed to vary for different atmospheric conditions (continuum to rarefied) to obtain a range of nozzle exit to ambient pressure ratios ($\frac{P_e}{P_\infty}$). The continuum simulation with atmospheric condition of $P_\infty = 1$ atm (1.01×10^5 Pa) will be used to verify the simulation results with reference results from [45]. Initial conditions are equilibrium states for both nozzle exit and ambient atmosphere, and equilibrium state temperature of ambient (T_∞) for all the simulations is 300K. Numerical simulation is executed for these range of pressure ratios for different nozzle exit Mach number. Reference Knudsen number for ambient atmosphere is obtained using:

$$\text{Kn}_{\text{ref}} = \frac{l_{\text{mfp}}}{D} \quad (28)$$

Nozzle exit diameter (D) of 15.875×10^{-3} m [30] is used as reference length and l_{mfp} is mean free path calculated using [57]:

$$l_{\text{mfp}} = \frac{k_B T_\infty}{\sqrt{2} \pi d^2 P_\infty} \quad (29)$$

where $k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann constant and d is diameter of Ar gas particle.

2. Parameter space for retro-rocket simulation

A rocket in the descent phase has a velocity that changes with deceleration. So, the plume flow from the rocket nozzle is subjected to some incoming velocity. To investigate the effects of retro-flow on the shock structures, 3 different flow speeds with an incoming mach number of M_{tr} are selected [58].

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Table 2 Parameter space for simulation in quiescent atmosphere

P_0 (Pa)	M_e	P_e (Pa)	P_∞ (Pa)	P_e/P_∞	Kn_{ref}
1.75×10^7	1.0	8.54×10^6	1.01×10^5	84.6	3.43×10^{-6}
			1.01×10^3	8.46×10^3	3.43×10^{-4}
			1.01×10^1	8.46×10^5	3.43×10^{-2}
			1.01×10^{-1}	8.46×10^7	3.43
1.75×10^7	2.0	2.11×10^6	1.01×10^5	20.8	3.34×10^{-6}
			1.01×10^3	2.08×10^3	3.34×10^{-4}
			1.01×10^1	2.08×10^5	3.34×10^{-2}
			1.01×10^{-1}	2.08×10^7	3.34
1.75×10^7	3.0	5.45×10^5	1.01×10^5	5.40	3.34×10^{-6}
			1.01×10^3	5.40×10^2	3.34×10^{-4}
			1.01×10^1	5.40×10^4	3.34×10^{-2}
			1.01×10^{-1}	5.40×10^6	3.34
1.75×10^7	4.0	1.72×10^5	1.01×10^5	1.71	3.34×10^{-6}
			1.01×10^3	1.71×10^2	3.34×10^{-4}
			1.01×10^1	1.71×10^4	3.34×10^{-2}
			1.01×10^{-1}	1.71×10^6	3.34
1.75×10^7	5.0	6.50×10^4	1.01×10^5	6.43×10^{-1}	3.34×10^{-6}
			1.01×10^3	64.3	3.34×10^{-4}
			1.01×10^1	6.43×10^3	3.34×10^{-2}
			1.01×10^{-1}	6.43×10^5	3.34

Table 3 Parameter space for retro-rocket simulation

M_{rtr}	P_0 (Pa)	M_e	P_e (Pa)	P_∞ (Pa)	P_e/P_∞	Kn_{ref}
M_{rtr}	1.75×10^7	3.0	5.45×10^5	1.01×10^5	5.40	3.34×10^{-6}
				1.01×10^3	5.40×10^2	3.34×10^{-4}
				1.01×10^1	5.40×10^4	3.34×10^{-2}
				1.01×10^{-1}	5.40×10^6	3.34

$$M_{\text{rtr}} = [2.4, 3.5, 4.6]$$

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